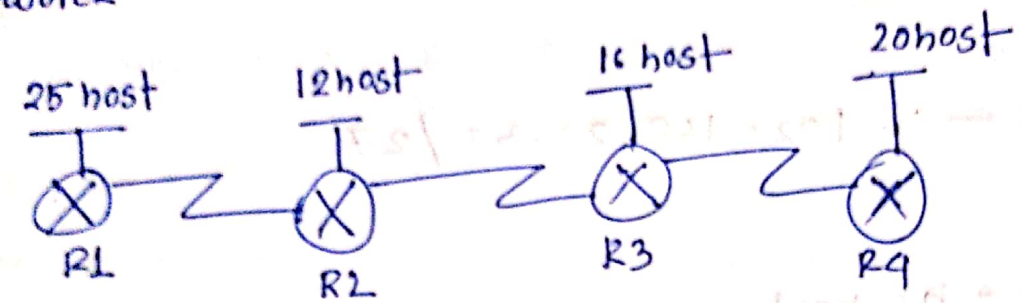
VLSM

1) Given network address is 192.168.20.0 and the original Subnet mask is 255.255.255.0. Consider the following network — (2017)



R1 — 25
R4 — 20
R3 — 16
R2 — 12

Step-1: R1 — 25 host

$2^5 - 2 = 30$

Subnet, $2^3 = 8$

difference $= 2^5 = 32$

(i) 192.168.20.0/27
(ii) 192.168.20.32/27
(iii) 192.168.20.64/27
⋮
(VIII) 192.168.20.224/27

∴ R1 → 192.168.20.0/27

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Step-2: R4 — 20 host

$2^5 - 2 = 30$

Subnet $2^3 = 8$

difference $2^5 = 32$

(i) 192.168.20.32/27

(ii) 192.168.20.64/27

⋮

R4 — 192.168.20.32/27

Step-3: R3 → 46 host

$2^5 - 2 = 30$

$2^3 = 8$, subnet

difference = $2^5 = 32$

(i) 192.168.20.64/27

(ii) 192.168.20.96/27

⋮

Step-4: R2 — 12 host

$2^4 - 2 = 14$

Subnet = $2^1 = 2$

difference = $2^4 = 16$

(i) 192.168.20.96/28

(ii) 192.168.20.112/28

⋮

Step-5: Now allocate address for each WAN. Each WAN requires 2 hosts.

range, $2^2 - 2 = 2$

Subnet, $2^2 = 4$

difference, $2^2 = 4$

(i) 192.168.20.112/30

(ii) 192.168.20.116/30

(iii) 192.168.20.120/30

(iv) 192.168.20.124/30

The available address for the WAN links can be taken from available address in each of the /30 subnets.

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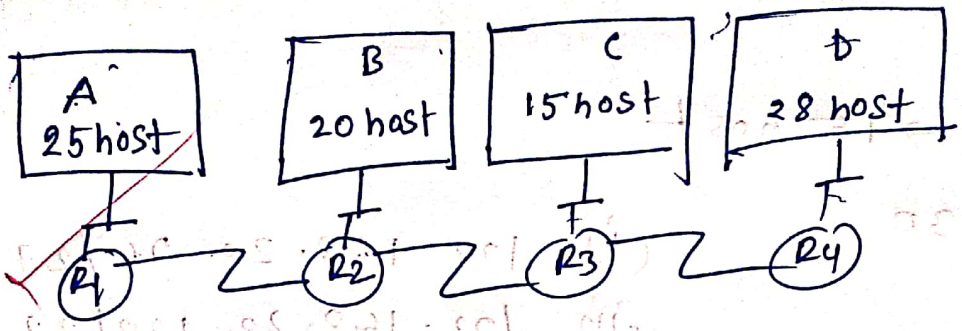
2)



(2015)

Given network address is 192.168.20.0/24. Perform VLSM.

D — 28
A — 25
B — 20
C — 15

Step-1: D — 28 host

range, $2^5 - 2 = 30$

Subnet, $2^3 = 8$

difference, $2^5 = 32$

(i) 192.168.20.0/27
(ii) 192.168.20.32/27
(iii) 192.168.20.64/27
⋮

∴ D — 192.168.20.0/27

Step-2: A — 25 host

$2^5 - 2 = 30$

sub: $2^3 = 8$

diff: $2^5 = 32$

(i) 192.168.20.32/27
(ii) 192.168.20.64/27
(iii) 192.168.20.96/27
⋮

A — 192.168.20.32/27

Step-3:

B — 20 host

range: $2^5 - 2 = 30$

Sub: $2^3 = 8$

diff: $2^5 = 32$

(i) 192.168.20.64/27
(ii) 192.168.20.96/27
⋮

B — 192.168.20.64/27

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Step-4:

C → 15 host

$2^5 - 2 = 30$

$2^2 = 8$

$2^5 = 32$

(i) 192.168.20.96/27

(ii) 192.168.20.128/27

C → 192.168.20.96/27

Step-5:

$2^2 - 2 = 2$

Sub: $2^3 = 8$

diff: $2^2 = 4$

(i) 192.168.20.128/30

(ii) 192.168.20.132/30

(iii) 192.168.20.136/30

(iv) 192.168.20.140/30

(2) 22

$2^7 - 2 = 6$

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